

- 1. Prerequisites** A knowledge of the specifications for P1, P2, P3, P4 and FP1, their prerequisites and associated formulae, is assumed and may be tested.
- 2. Examination**
- First assessment: June 2020.
 - The assessment is 1 hour and 30 minutes.
 - The assessment is out of 75 marks.
 - Students must answer all questions.
 - Calculators may be used in the examination. Please see *Appendix 6: Use of calculators*.
 - The booklet *Mathematical Formulae and Statistical Tables* will be provided for use in the assessments.
- 3. Notation** Students will be expected to understand the symbols outlined in *Appendix 7: Notation*.
-

FP3.3 Unit content

What students need to learn:		Guidance
1. Hyperbolic functions		
1.1	Definition of the six hyperbolic functions in terms of exponentials. Graphs and properties of the hyperbolic functions.	For example, $\cosh x = \frac{1}{2}(e^x + e^{-x})$, $\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}.$ Students should be able to derive and use simple identities such as $\cosh^2 x - \sinh^2 x \equiv 1 \text{ and}$ $\cosh^2 x + \sinh^2 x \equiv \cosh 2x$ and to solve equations such as $a \cosh x + b \sinh x = c$.
1.2	Inverse hyperbolic functions, their graphs, properties and logarithmic equivalents.	For example, $\operatorname{arsinh} x = \ln \left[x + \sqrt{1 + x^2} \right]$. Students may be required to prove this and similar results.